

Varun Shankar

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Salt Lake City, UT 84112, USA *url:* http://users.cs.utah.edu/~shankar

RESEARCH INTERESTS Scientific machine learning, scientific computing, high performance computing.

CURRENT EMPLOYMENT Assistant Professor, Kahlert School of Computing, University of Utah, **July 2023 - Present**.
Associate Chair, FinTech Center, University of Utah, **July 2023 - Present**.

OTHER ROLES Adjunct Graduate Faculty, Dept. of Mathematics, Boise State University, **April 2019 - Present**.
Advisor, Light-Hope Diagnostics, Inc., **July 2026 - Present**.

PAST EMPLOYMENT Associate Director, Master of Software Development Program, University of Utah, **July 2022 - September 2026**. Asst. Prof. Lecturer, Kahlert School of Computing, University of Utah, **July 2018 - June 2023**.
Co-interim Director, Master of Software Development, University of Utah, **July 2022 - June 2023**.
Postdoctoral position, Dept. of Mathematics, University of Utah, **July 2014 - June 2018**.

EDUCATION **University of Utah**, Salt Lake City, Utah USA, **2009-2014**
PhD. in Computing, Scientific Computing.

University of California, Davis, California USA, **Jan 2009-March 2009**
Visiting Scholar, UC Davis Global Study Program.

SRM University, Chennai, Tamil Nadu India, **2005-2009**
Bachelor of Technology, Computer Science and Engineering.

GRANTS

Agency	Total Amount (\$)	Utah/My Amount (\$)	Role	Details
NSF Future Core RI	778k	778k/389k	PI	In review, sub. Sep 2026
NSF Career	670k	670k/670k	PI	In review, sub July 2026
DoE CS2	100k	100k/33k	Co-PI	Funded, active, 2026-2029
NSF CS2	385k	385k/128k	Co-PI	Funded, active, 2026-2029
NSF FDT BioTech	614k	414k/414k	PI	Declined, sub 2026
NSF CAIG	1.03M	1.03M/343k	Co-PI	Declined, sub 2026.
AFRL/ARCTOS	213k	213k/213k	PI	Funded, active, 2025-2026.
NSF CDS&E DMS	~450k	~250k/200k	PI	Funded, active, 2025-2028.
NSF MFAI	1.2 M	800k/400k	Co-PI	Declined, sub. 2024.
AFRL	500k	500k/410k	PI	Funded yearly, active, 2023-2027.
AFOSR (LRIR)	~539k	481k/466k	PI	Funded, active, 2024-2027.
NSF CISE/CCF SHF	819k	819k/273k	Co-PI	Funded, active, 2024-2027.
Sandia (LDRD)	300k	300k	Co-PI	Funded, active, 2024-2027.
NSF DMS FRGMS	1.5 j	980k/~275k	Co-PI	Declined, sub. 2023.
NSF CISE OAC Core	~600k	600k/300k	PI	Declined, sub. 2023.
DoE ASCR	6 million	900k/300k	Co-PI	Declined, sub. 2023.
AFRL/NIAR	250k	250k/50k	Co-PI	Funded, 2022-2024.
NSF CISE AF Small	~323k	88k/88k	Utah PI	Funded, 2017-2022.
NSF CDS&E-DMS	450k	450k/150k	Co-PI	Funded, 2015-2020.

HONORS AND AWARDS

- College of Engineering Dean’s teaching recognition, 2022-2023.
- Nominated for School of Computing Outstanding Teacher award, 2022.
- College of Engineering Dean’s teaching recognition, 2021-2022.
- College of Engineering Dean’s teaching recognition, 2020-2021.
- NSF travel award for PNWNAS 2012.
- Bharat Petroleum Corporation Limited (BPCL) Scholarship for higher studies abroad (2010).
- TataChem Foundation Scholarship award for higher studies abroad (2009).
- SRM University Merit scholarship 2007-2008 for being in the top 2% of my batch.
- SRM University Merit scholarship 2006-2007 for being in the top 2% of my batch.

PUBLICATIONS

Google Scholar Page: <https://scholar.google.com/citations?user=goYJXTsAAAAJ&hl=en>

1. Branden Frieden, Ryan Whitehead, Keith Ballard, Robert M. Kirby, and **Varun Shankar**. The Frame Kernel Method for Multiscale Operator Learning (In review, 2026).
2. Matthew Lowery, Grady B. Wright, and **Varun Shankar**. A high-order, meshless, Lagrangian–Eulerian RBF-FD method for advection–diffusion–reaction on moving manifolds (In review, 2026).
3. Branden Frieden, James M Ferguson, Alan Kuntz, and **Varun Shankar**. Neural Operators for Design-Space Surrogate Modeling of Tendon-Actuated Continuum Robots (IEEE International Conference on Robotics and Automation, ICRA 2026).
4. Matthew Lowery, John Turnage, Zachary Morrow, John D Jakeman, Akil Narayan, Shandian Zhe, and **Varun Shankar**. Kernel Neural Operators (KNOs) for Scalable, Memory-efficient, Geometrically-flexible Operator Learning (Transactions on Machine Learning Research, 2026).
5. Madison Cooley, Shandian Zhe, Robert M Kirby, and **Varun Shankar**. Polynomial-Augmented Neural Networks (PANNs) with Weak Orthogonality Constraints for Enhanced Function and PDE Approximation (SIAM Journal on Scientific Computing, 2026).
6. Madison Cooley, Ramansh Sharma, Shandian Zhe, Robert M. Kirby, and Varun Shankar. Localized Operator Learning with Adaptive Partition-of-Unity Mixture-of-Expert Networks (In review, 2026).
7. Ramansh Sharma, Matthew Lowery, Houman Owhadi, and Varun Shankar. Fluids You Can Trust: Property-Preserving Operator Learning for Incompressible Flows (In review, 2026).
8. Matthew Lowery, Zhitong Xu, Da Long, Keyan Chen, Daniel S. Johnson, Yang Bai, **Varun Shankar**, and Shandian Zhe. Deep Gaussian Processes for Functional Maps (In review, 2026).
9. Ben Charoenwong, Pratik Soni, **Varun Shankar**, Robert M. Kirby, and Jonathan Reiter. Blockchain Compliance: A Framework for Evaluating Regulatory Approaches (In review, 2025).
10. John Turnage, Matthew Lowery, John Jakeman, Zackary Morrow, Akil Narayan, and Varun Shankar. An Optimal Weighted Least-Squares Method for Operator Learning (In review, 2025).
11. Milena Belianovich, Gregory E. Fasshauer, Akil Narayan, and **Varun Shankar**. A Unified Framework for Efficient Kernel and Polynomial Interpolation (Journal of Approximation Software, 2026).
12. Alberto Cattaneo, **Varun Shankar**, and Michael Ballard. Rootfinding and Optimization Techniques for Solving Nonlinear Systems of Equations Arising from Cohesive Zone Models (AIAA SciTech 2025 Forum).
13. Ramansh Sharma and **Varun Shankar**. Ensemble and Mixture-of-Experts DeepONets For Operator Learning (Transactions on Machine Learning Research, 2025).
14. Madison Cooley, Robert M. Kirby, Shandian Zhe, and **Varun Shankar**. HyResPINNs: A Hybrid Residual Physics-Informed Neural Network Architecture Designed to Balance Expressiveness and Trainability (Transactions on Machine Learning Research, 2025).

15. Madison Cooley, **Varun Shankar**, Robert M. Kirby, and Shandian Zhe. Fourier PINNs: From Strong Boundary Conditions to Adaptive Fourier Bases (Transactions on Machine Learning Research, 2024).
16. Urban Duh, **Varun Shankar**, and Gregor Kosec. Discretization of non-uniform rational B-spline (NURBS) models for meshless isogeometric analysis (Journal of Scientific Computing, 2024).
17. Mingxuan Han, Chenglong Ye, **Varun Shankar**, and Jeff Phillips. Locally Adaptive and Differentiable Regression (Journal of Machine Learning for Modeling and Computing, 2023).
18. Grady B. Wright, Andrew M. Jones, and **Varun Shankar**. MGM: A meshfree geometric multilevel method for systems arising from elliptic equations on point cloud surfaces (SIAM Journal on Scientific Computing (SISC), 2023).
19. Ramansh Sharma and **Varun Shankar**. Accelerated training of physics-informed neural networks (PINNs) using Meshless Discretizations (NeurIPS 2022).
20. Keith M. Ballard, Endel Iarve, John Whitcomb, **Varun Shankar**, Hari Sundar, Lauren Ferguson, Robert M. Kirby, and David Mollenhauer. Remaining Challenges in the Application of High-Performance Computing to a Process-To-Performance Pipeline for Textile Composites (Proceedings of the American Society for Composites-Thirty-Seventh Technical Conference, September 2022).
21. Andrew Kassen, Aaron Barrett, **Varun Shankar**, and Aaron L. Fogelson. Immersed boundary simulations of cell-cell interactions in whole blood (Journal of Computational Physics, August 2022).
22. Andrew Kassen, **Varun Shankar**, and Aaron L. Fogelson. A fine-grained parallelization of the Immersed Boundary method (International Journal of High Performance Computing Applications (IJHPCA), June 2022).
23. Keith M. Ballard, Roman Amici, **Varun Shankar**, Lauren A. Ferguson, M. Braginsky, and Robert M. Kirby. Towards an Extrinsic, CG-XFEM approach based on hierarchical enrichments for modeling progressive fracture (Computer Methods in Applied Mechanics and Engineering (CMAME), January 2022).
24. **Varun Shankar**, Grady B. Wright, and Aaron Fogelson. An efficient high-order meshless method for advection-diffusion equations on time-varying irregular domains (Journal of Computational Physics, August 2021).
25. Hallie Elich, Aaron Barrett, **Varun Shankar**, Aaron L. Fogelson. Pump efficacy in a two-dimensional, fluid-structure-interaction model of a chain of contracting lymphangions (Biomechanics and Modeling in Mechanobiology, July 2021).
26. **Varun Shankar**, Grady B. Wright, and Akil Narayan. A Robust Hyperviscosity Formulation for Stable RBF-FD Discretizations of Advection-Diffusion-Reaction Equations on Manifolds (SIAM Journal on Scientific Computing, August 2020).
27. Sean Lawley and **Varun Shankar**. Asymptotic and numerical analysis of a stochastic PDE model of volume transmission (SIAM Multiscale Modeling and Simulation, May 2020).
28. Keith M. Ballard, Roman Amici, **Varun Shankar**, and Robert M. Kirby. A preliminary assessment of a new XFEM framework for predicting complex fracture (22nd International Conference on Composite Materials (ICCM22), August 2019).
29. **Varun Shankar** and Aaron L. Fogelson. Hyperviscosity-based Stabilization for Radial Basis Function-Finite Difference (RBF-FD) discretizations of advection-diffusion equations (Journal of Computational Physics, November 2018).
30. **Varun Shankar**, Akil Narayan, and Robert M. Kirby. RBF-LOI: Augmenting Radial Basis Functions (RBFs) with Least Orthogonal Interpolants (LOI) for Solving PDEs on Surfaces (Journal of Computational Physics, November 2018).

31. **Varun Shankar**, Aaron L. Fogelson and Robert M. Kirby. Robust Node Generation for Mesh-free Discretizations on Irregular Domains and Surfaces (SIAM Journal on Scientific Computing, August 2018).
32. **Varun Shankar** and Grady B. Wright. Mesh-free Semi-Lagrangian Methods for Advection on a Sphere using Radial Basis Functions (Journal of Computational Physics, August 2018).
33. Santosh Pokhrel, **Varun Shankar**, and Jamesina J. Simpson. 3-D FDTD Modeling of Electromagnetic Wave Propagation in Magnetized Plasma Requiring Singular Updates to the Current Density Equation (IEEE Transactions on Antennas and Propagation, June 2018).
34. Vidhi Zala, **Varun Shankar**, Shankar P. Sastry, and Robert M. Kirby. Curvilinear Mesh Rectification using Radial Basis Function Interpolation and Smoothing (Journal of Scientific Computing, April 2018).
35. Santosh Pokhrel, **Varun Shankar**, and Jamesina J. Simpson. Simplified FDTD model of electromagnetic wave propagation in magnetized plasma (Applied Computational Electromagnetic Society Symposium [ACES], March 2018).
36. **Varun Shankar**. The Overlapped Radial Basis Function-Finite Difference (RBF-FD) Method: A Generalization of RBF-FD (Journal of Computational Physics, August 2017).
37. E. Lehto, **Varun Shankar** and Grady B. Wright. A Radial Basis Function (RBF)-Based Compact Finite Difference (FD) Scheme for Reaction-Diffusion Equations on Surfaces (SIAM Journal on Scientific Computing, September 2017).
38. Edward J. Fuselier, **Varun Shankar** and Grady B. Wright. A High-Order Radial Basis Function (RBF)-Based Leray Projection Method for the Incompressible Stokes Equations (Computers and Fluids, April 2016).
39. **Varun Shankar**, Grady B. Wright, Robert M. Kirby, and Aaron L. Fogelson. Augmenting the Immersed Boundary Method with Radial Basis Functions (RBFs) for the modeling of platelets in hemodynamic flows (Int. J. Numer. Meth. Fluids, July 2015).
40. **Varun Shankar** and Sarah D. Olson. Radial Basis Function (RBF)-Based Parametric Models for Open and Closed Curves within the Method of Regularized Stokeslets (Int. J. Numer. Meth. Fluids, May 2015).
41. **Varun Shankar**, Grady B. Wright, Robert M. Kirby, and Aaron L. Fogelson. A Radial Basis Function (RBF)-Finite Difference Method for Diffusion and Reaction-Diffusion Equations on surfaces (Journal of Scientific Computing, September 2014).
42. **Varun Shankar**, Grady B. Wright, Aaron L. Fogelson, and Robert M. Kirby. A Radial Basis Function (RBF)-Finite Difference Method for the Simulation of Reaction-Diffusion Equations on Stationary Platelets within the Augmented Forcing Method (Int. J. Numer. Meth. Fluids, January 2014).
43. **Varun Shankar**, Grady B. Wright, Aaron L. Fogelson, and Robert M. Kirby, A study of different modeling choices for simulating platelets within the immersed boundary method (Applied Numerical Mathematics, January 2013).

PHD STUDENTS

1. Ramansh Sharma, KSoC PhD student.
2. Alberto Cattaneo, KSoC PhD student.
3. Matthew Lowery (Chair: Shankar, Co-Chair: Shandian Zhe), KSoC PhD student.
4. Branden Frieden, KSoC PhD student (former MSD student).
5. Andrey Pluzhnik, KSoC PhD student.

PHD COMMITTEES	1. Anh Tran, KSoC (advised by Ganesh Gopalakrishnan). 2. Filip Belik, Math (advised by Akil Narayan and Christel Hohnegger). 3. John Turnage, Math (advised by Akil Narayan).
FORMER MENTEES	1. Dr. Madison Cooley, KSoC/SCI PhD student (Chair: Robert M. Kirby, Co-Chair: Varun Shankar). Graduated 2025. 2. Milena Belianovich, KSoC MS student. Graduated 2025. 3. Dr. Andrew Jones, Department of Mathematics, Boise State University (co-advised with Grady B. Wright). Now at Sandia National Laboratory. Graduated 2024. 4. Dr. Andrew Kassen, Department of Mathematics (co-advised with Aaron L. Fogelson). Now at Intel. Graduated 2021. 5. Dr. Hallie Elich, Department of Mathematics (co-advised with Aaron L. Fogelson). Now at L3Harris. Graduated 2021. 6. Hannah Bruns, undergraduate thesis, SoC. UROP award recipient. Now at BioFire. 7. Zixuan Zhao, MSD graduate, KSoC. Hourly research assistant. 8. Roman Amici, MS student, KSoC (co-advised with Robert M. Kirby). 9. Sage Shaw, MS student, Department of Mathematics, Boise State University (co-advised with Grady B. Wright). Currently PhD student at CU Boulder. 10. Daniel Malmuth, MS student, Department of Mathematics, Boise State University (co-advised with Grady B. Wright). 11. M.S. Srivatsa, M.S. in Computer Science, KSoC (co-advised with Robert M. Kirby). Currently at Microsoft. 12. Harshitha Parnandi, M.S. in Computer Science, KSoC (co-advised with Robert M. Kirby). 13. Dr. Vidhi Zala, during her M.S. in Computer Science, School of Computing (co-advised with Robert M. Kirby). Currently at Intel.
INTERNAL SERVICE	1. Reviewer, AI Moonshot proposals, September 2026. University-level service. 2. Member, Technology Commercialization Advisory Committee, August 2026 - present. University-level service 3. Faculty Coordinator for Web & Social Media Engagement, Kahlert School of Computing, Aug 2026 - present. 4. Associate Chair, FinTech Center, July 2023- present. University-level service. 5. Director, Scientific Computing Track, July 2023- present. 6. Associate Director, Master of Software Development (MSD) program, July 2022- September 2026. 7. Head of lecturing faculty hiring committee, Spring 2023. 8. KSoC graduate admissions committee, 2018-present. 9. MSD admissions committee, 2018-present. 10. MSD industry seminar, Fall and Spring 2022- present. 11. MSD capstone committee, 2018-present. 12. MSD steering committee, 2018-present. 13. KSoC Grad Visit, Spring 2022. 14. KSoC PhD recruiting event, Fall 2021. 15. MSD lab instruction/assistance, Fall 2021.

16. Hiring committee for Brent Stephens, Fall 2020.
17. Co-organizer of poster session for SoC Grad Visit, Spring 2020 (with Marina Kogan).
18. MSD outreach: Silicon Slopes talks, booth. Fall 2018, Spring 2019.
19. MSD capstone committee, Fall 2018.

EXTERNAL SERVICE

1. Organizer (with Oleg Davydov) of the 4-part minisymposium titled “Meshless Numerical Methods and Scientific Machine Learning with Kernels and Gaussian Processes” at the International Conference on Spectral and High-Order Methods (ICOSAHOM), July 2027. 16 speakers.
2. Organizer (with postdoc Nathan Willis) of the 2-part minisymposium titled “Automatic Differentiation for Scientific Computing” at the SIAM Conference on Computational Science and Engineering, February 2027. 8 speakers.
3. Organizer (with Grady B. Wright) of the 2-part minisymposium titled “Kernel methods for PDEs and ML” at the SIAM Conference on Mathematics of Data Science, November 2026. 8 speakers.
4. Editor (by invitation), Nature Scientific Reports, September 2026 - present.
5. Chair, Panel Discussion on kernel approximation and Gaussian process software: Libraries, Workflows, and Best Practices, Banff Workshop on Kernel Approximation and Gaussian Processes, September 2026.
6. Local Arrangements Chair for International Conference on Supercomputing (ICS) 2025.
7. FinTech Xchange Summit 2025, Organizing Committee.
8. FinTech Xchange Summit 2024, Organizing Committee.
9. Organizer (with Grady B. Wright) of the 2-part minisymposium titled “Recent advances in Kernel-based Meshless Methods” at the conference Approximation Theory 16, May 19-22, 2019. This minisymposium brought together leading theoreticians and practitioners in the world of meshless methods. 8 speakers.
10. Head organizer for poster session, member of Local Organizing Committee, Society of Mathematical Biology Annual meeting (<http://www.smg.org>), University of Utah, July 17-20, 2017. 100+ posters.
11. Co-organizer of the conference “Modeling Complex Fluids and Gels for Biological Applications”, University of Utah, May 4-6, 2017. This conference brought together leading researchers in the field of complex biofluids. 30 speakers.
12. Co-organizer (with Grady B. Wright) of the 5-part minisymposium titled “Advances in Radial Basis Function and Other Meshfree Methods”, SIAM CSE 2015. This minisymposium had 20 of the top researchers in meshfree methods from across the world as speakers. 20 speakers.

CONFERENCE TALKS AND OTHER PRESENTATIONS

1. “Interpretable Multiscale Operator Learning with Multiscale Kernel Frames”, Banff Workshop on Kernel Approximation and Gaussian Processes, September 2026. Invited by Grady Wright, Thomas Hangelbroek, Christian Rieger, Cecile Piret, and Janin Jeagar.
2. “Multiscale Operator Learning with Multiscale Kernel Frames”, World Congress on Computational Mechanics with the European Community on Computational Methods in Applied Sciences (WCCM-ECCOMAS), July 2026. Invited by Rüdiger Kempf.
3. “Property-Preserving and Multiscale Operator Learning for Surrogate Modeling”, Air Force Research Laboratory, June 2026. Invited by Keith Ballard.
4. “Property-Preserving and Multiscale Operator Learning for Surrogate Modeling”, Air Force Office of Scientific Research PI Meeting, December 2025.
5. “Kernels in Numerical Methods and Operator Learning”, CalTech Computational Mathematics + X (CMX) Seminar, April 2025. Invited by Houman Owahdi.

6. “Kernel Methods for Operator Learning”, SIAM Conference on Computational Science and Engineering, March 2025. Invited by Jonah Reeger.
7. “Kernel methods for Operator Learning”, Colorado School of Mines Applied Mathematics Seminar, December 2024. Invited by Gregory Fasshauer.
8. “Case Studies in Scientific Machine Learning: Philosophy, Applications, and Opportunities”. International Guest Lecture at SRM University, April 2023. Invited by R. Angeline.
9. “Physics-informed neural networks (PINNs) and kernel methods”, SIAM Conference on Computational Science and Engineering, Session on Meshless Methods, February 2023. Invited by Elisabeth Larsson.
10. “Accelerated training of physics-informed neural networks (PINNs) using meshless discretizations”, NeurIPS 2022.
11. “Accelerated training of physics-informed neural networks (PINNs) using meshless discretizations”, CRUNCH seminar, Brown University, August 2022. Invited by George Karniadakis.
12. “Stacked surface representations for Geometric Modeling and Node Generation”, SIAM Conference on Geometric and Physical Modeling, Session on Meshless Methods, September 2021. Invited by Grady Wright.
13. “A High-Order Meshless Semi-Lagrangian Method for PDEs on Surfaces”, Dolomites Workshop on Constructive Approximation and Applications (dedicated to Robert Schaback on his 75th birthday), Session on Meshless Methods, September 2021. Invited by Gabriele Santin and Emma Perrachione.
14. “A High-Order Meshless Semi-Lagrangian Method for PDEs on Surfaces”, SIAM Conference on Mathematical and Computational Issues in the GeoSciences, Minisymposium on Approximation in Geosciences: Theory and Numerics, June 2021. Invited by Alessandra de Rossi and Francesco Dell’Accio.
15. “Case Studies in Scientific Computing: Philosophy, Applications, and Opportunities”. International Guest Lecture at SRM University, Ramapuram Campus, April 2021. Invited by R. Angeline and K. Raja.
16. “An efficient high-order meshless method for Advection-Diffusion Equations on Time-varying Irregular Domains”, Online Seminars on Numerical Approximation and Applications (OSNA²), November 2020. Invited by Victor Bayona, Costanza Conti, Emma Perracchione, and Rosanna Campagna.
17. “An efficient high-order meshless method for Advection-Diffusion Equations on Time-varying Irregular Domains”, Approximation Theory and Numerical Analysis, American Mathematical Society (AMS) Sectional Meeting, October 2020. Invited by Akil Narayan and Vera Babenko.
18. “Automatic Hyperviscosity-based Stabilization of RBF-FD Discretizations”, Approximation Theory 16, session on meshless methods, May 2019.
19. “RBF-FD for PDEs on Manifolds: Stable and Stagnation-free Formulations”, SIAM Conference on Computational Science and Engineering Minisymposium on Meshfree Methods: Computational Advances and Applications, March 2019. Invited by Grady Wright.
20. “A high-order meshfree framework for solving PDEs on irregular domains and surfaces ”, Colorado School of Mines, April 2018. Invited by Greg Fasshauer.
21. “A high-order meshfree framework for solving PDEs on irregular domains and surfaces ”, Boise State University, February 2018. Invited by Grady Wright.
22. “A high-order meshfree framework for solving PDEs on irregular domains and surfaces ”, University of Massachusetts Dartmouth, January 2018. Invited by Sigal Gottlieb.

23. “KernelPack: A C/C++ library for high-order RBF-FD methods and automatic node generation on irregular domains”, Software Session, ICERM Topical Workshop on Localized Kernel-Based Meshless Methods for Partial Differential Equations, Brown University, August 2017. Invited by Natasha Flyer, Oleg Davydov, Bengt Fornberg, and Elisabeth Larsson.
24. “Efficient High-Order RBF-FD methods for Advection-Diffusion-Reaction Equations on Irregular Domains”, ICERM Topical Workshop on Localized Kernel-Based Meshless Methods for Partial Differential Equations, Brown University, August 2017. Invited by Natasha Flyer, Oleg Davydov, Bengt Fornberg, and Elisabeth Larsson.
25. “A high-order meshfree framework for solving PDEs on irregular domains and surfaces ”, Colorado School of Mines, June 2017. Invited by Greg Fasshauer.
26. “A high-order meshfree framework for solving PDEs on irregular domains and surfaces ”, National Center for Atmospheric Research, May 2017. Invited by Natasha Flyer.
27. “A Meshfree Framework for Solving PDEs on arbitrary domains and surfaces”, Modeling Complex Fluids and Gels for Biological Applications. Organizer, May 2017.
28. “The Overlapped RBF-FD Method”, SIAM CSE 2017, Minisymposium on Meshfree Methods. Invited by Nathaniel Trask.
29. “Radial Basis Function (RBF)-Based Meshfree Methods for Advection-Diffusion-Reaction PDEs on Arbitrary Surfaces”, SIAM Conference on Mathematical Aspects of Material Science Minisymposium on Particle Methods, May 2016. Invited by Nathaniel Trask.
30. “Radial Basis Function (RBF)-Based Meshfree Methods for Transport on the Sphere and Other Surfaces”, 15th International Conference on Approximation Theory, May 2016. Invited by Edward Fuselier.
31. “Radial Basis Function (RBF)-Based Meshfree Methods for Transport on the Sphere”, Boise State University, March 2016. Invited by Grady Wright.
32. “A Radial Basis Function (RBF)-Based Leray Projection Method for the Incompressible Unsteady Stokes Equations”, Applied Math Seminar, Arizona State University, September 2015. Invited by Rodrigo Platte.
33. “A Radial Basis Function (RBF)-Based Leray Projection Method for the Incompressible Unsteady Stokes Equations”, SIAM CSE 2015, Minisymposium on Meshfree Methods. Organizer.
34. “RBF-FD for the solution of Diffusion and Reaction-Diffusion Equations on surfaces”, International Conference on Spectral and High-Order Methods (ICOSAHOM), June 24, 2014. Invited by Rodrigo Platte.
35. PhD Thesis Defense. “Radial Basis Function-Based Numerical Methods for the Simulation of Platelet Aggregation”, April 18, 2014.
36. “Radial Basis Functions for the simulation of platelet aggregation”, Applied Mathematics Seminar, Department of Mathematics, University of Utah, October 28, 2013.
37. “RBF-FD for the solution of Reaction-Diffusion Equations on surfaces in $\mathbb{R}^3$ ”, Fluids/Gel Seminar, Department of Mathematics, University of Utah, September 11, 2013.
38. “RBF-FD for reaction-diffusion equations on manifolds and RBF symmetric Hermite interpolation-based modifications to forcing methods”, SIAM Annual Meeting Minisymposium on “Advances in Radial Basis Function and Other Meshfree Methods”, July 9, 2013. Invited by Edward Fuse-lier.
39. “Using RBFs within the Immersed Boundary method”, SIAM Conference on Computational Science and Engineering Minisymposium on “Recent Advances in Immersed Boundary Meth-ods”, February 25, 2013. Invited by Boyce Griffith and Robert Guy.
40. (Poster) “Using RBFs within the Immersed Boundary method”, 25th Annual Pacific Northwest Numerical Analysis Seminar, October 27, 2012. Invited by Grady Wright, Donna Calhoun and Jodi Mead.

41. “RBF methods for chemical transport in clotting”, June 17, 2012, Graphics Seminar, School of Computing, University of Utah.
42. “An RBF Immersed Boundary Method”, December 2, 2011, Fluids/Gel Seminar, Department of Mathematics, University of Utah.
43. “A CUDA (GPGPU) Implementation of the 2D Immersed Boundary Method”, April 11, 2010, Fluids/Gel Seminar, Department of Mathematics, University of Utah.
44. “The Immersed Boundary Method”, March 2, 2010, Graphics Seminar, School of Computing, University of Utah.

REVIEWING
EXPERIENCE

Reviewed for

1. NeurIPS.
2. ICLR.
3. Neurocomputing
4. SIAM Journal on Scientific Computing (SISC).
5. Journal of Computational Physics (JCP).
6. International Journal for Numerical Methods in Engineering (IJNME).
7. Journal of Scientific Computing (JSC).
8. Applied Numerical Mathematics (APNUM).
9. Computers and Fluids.
10. Computers and Mathematics with Applications.
11. Engineering Analysis with Boundary Elements.
12. BIT Numerical Mathematics.

TEACHING
EXPERIENCE

Instructor

Algorithms (CS 4150), Fall 2025.
 Mobile Application Programming (CS 4530), Fall 2024.
 CS 6220/COMP 5960: Advanced Scientific and Data Computing II, Spring 2024.
 MSD: Introduction to Software Development (CS 6010), Fall 2022-2023.
 Mobile Application Programming (CS 4530), Spring 2022-2023.
 MSD: Systems I (CS 6013), Spring 2020-2023.
 MSD: Computer Programming (CS 6011), Fall 2022.
 MSD: Data Structures and Algorithms (CS 6012), Fall 2018-2021.
 MSD: Application System Design (CS 6018), Fall 2018-2021.
 MSD: Database Systems (CS 6016), Summer 2019-2020.
 Intro to Scientific and Data Computing (CS 3200), Spring 2019.
 MSD: Data Structures and Algorithms (CS 6012), Fall 2018.
 Numerical Analysis II (Math 5620/6865), Spring 2016-2017.
 Differential Equations and Linear Algebra (Math 2250), Spring 2015, Fall 2016.
 Vector Calculus and PDEs for Engineers (Math 3140), Fall 2015.
 Engineering Calculus II (Math 1320), Fall 2014.

Teaching Assistant

Advanced Scientific Computing II, Spring 2013-2014.
 Advanced Scientific Computing I, Fall 2012-2013.

COURSE DESIGN	I developed CS 6018, Application System Design, which is the final course in the Master's in Software Development (MSD) program in the School of Computing, University of Utah. The course is focused on developing applications for Android devices, using this platform as a tool to teach good software design and engineering. Over Summer 2022, I updated this class to use Kotlin rather than Java to keep up with industry standards. I continue to update this class with Professor Ben Jones. We also update CS 4530 jointly every year. Over Spring 2020-2021, I updated CS 6013 (MSD) to incorporate GPU and multicore programming. Math 5620/6865 required me to develop my own syllabus and course structure. I have developed a simple yet unifying technique for teaching Numerical Analysis of ODEs and PDEs, and my own typeset course materials. I've simplified this material, updated it to include numerical linear algebra and data science techniques, and adapted it to CS 3200 and now to CS 6220/COMP 5960.
PROFESSIONAL SOCIETIES	I am currently a member of the Society for Industrial and Applied Mathematics (SIAM) and the American Mathematical Society (AMS).